

Modal Energy Injection in Three Fixed-Budget Contact Solvers: Measurement and a Cumulative Storage Bound

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Abstract

Unilateral contact between reduced (modal) deformable bodies and rigid bodies is commonly resolved by a velocity-level complementarity law in which one shared multiplier couples the rigid and modal unknowns. Interactive solvers evaluate this law under small, fixed local iteration budgets; we measure the energetic cost of that truncation. Running one transcribed contact row through three fixed-budget solver formulations — position-based (XPBD), augmented-Lagrangian (AVBD), and an implicit sequential-impulse realization — over an identical 24-cell scene $\times$ relaxation $\times$ budget sweep, we observe behaviour that differs sharply across the three tested implementations. Measured against the gross rigid-side loss (a scene-wide, gravity-corrected quantity, not a per-contact dissipation), the position-based host overdraws by up to 4.4×10^7 J — 1.2×10^5 times the incident rigid kinetic energy in the most starved cells — the augmented-Lagrangian host in 3 of 24 cells by at most 6.7 J, and the implicit realization never: a catastrophic pathology beside a small and localized one, five orders of magnitude apart, which the customary ratio-to-incident-energy diagnostic compresses into a single number. An iteration-budget sweep identifies the position-based amplification as a truncation artifact, decaying toward an implicit high-iteration reference computed in the same code path, closing the gap by six orders of magnitude. Since that regime is unaffordable at interactive budgets we add a deliberately minimal fail-safe: a cumulative modal-storage bound funded by the measured gross rigid-side loss, maintained by a reservoir ledger and a projection of the realized modal state, whose cost we measure rather than assume. The loop guarantees that bound unconditionally — for any host and any contact solve, up to a stated accounting tolerance — and the realized margins sit at the roundoff floor (1.1×10^{-13} J) in all 90 measured cells (78 distinct configurations), including the 1×8 and 2×4 budgets at which interactive solvers are commonly deployed; what it costs is contact validity: worst-case post-projection penetration 21.6 mm (72% of the board's thickness) at the adversarial 4×1 corner, 9.8 mm at the deployed budgets.

CCS Concepts

• Computing methodologies → Physical simulation.

Keywords

contact, modal reduction, passivity, position-based dynamics

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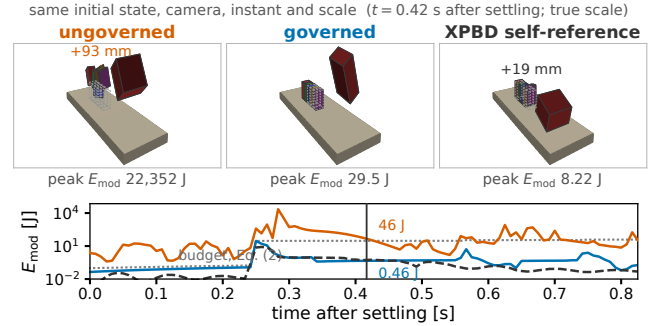


Figure 1: One shelf impact at a production-like interactive budget (1×8), true scale, three independent runs. Ungoverned, truncation energy throws the resting books up to 93 mm — more than their own height; governed, (2) holds peak modal energy to 29.5 J and they stay within 1 mm of rest; the reference is the host's own high-iteration self-reference (500×1). The bound removes the spurious launch and some legitimate motion with it: the reference lifts one book 19 mm, the governed run none.

1 Introduction

A deformable object that is struck at one point and rings elsewhere is routinely modelled by projecting the elastic response onto a small modal basis. When such a body also rests in unilateral contact with rigid bodies, the established treatment resolves rigid and modal unknowns together through a shared contact multiplier at the velocity level [17]. The one-way alternative — driving a separate modal filter from contact events without back-reaction — is the real-time precedent [3, 9]. Interactive modal deformation subject to collision and contact constraints predates both [8].

We take this coupled row as given. Written at position level it sits one linearization step from the velocity-level law: for a gap C , $C^{n+1} \approx C^n + h \mathbf{J} \mathbf{u}^{n+1}$, which divided by h recovers the velocity-level complementarity condition with a C^n/h stabilization bias, carrying the same Jacobian and multiplier as Zheng and James [17]. The row is *not* a contribution of this paper, and our transcription is narrower than the original law: modes couple through normal rows only, and the position-level form assumes $e = 0$ restitution.

Interactive solvers, however, do not solve this row to convergence: they run a fixed, small number of local iterations per substep, and it is known that finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [15] — an observation that likewise predates this work. Velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative

solvers named, goes back at least to Kaufman et al. [10]. What is missing is a quantitative account of this case — unilateral contact-to-modal transfer — and a mitigation that fits within an interactive budget.

Contributions.

- (1) **Measurement.** A single contact row evaluated in three fixed-budget solver implementations over one identical 24-cell adversarial sweep on one machine (§3.1), plus an iteration-budget sweep against an implicit high-iteration reference identifying the position-based injection as a truncation artifact (§3.2).
- (2) **A cumulative storage bound.** A one-directional bound on the energy entering modal storage, funded by the measured gross rigid-side loss and maintained by a reservoir ledger and a projection of the realized modal state (§2). It is the simplest admissible enforcement — a deliberately minimal, audited fail-safe, not a constructive contact method — and it guarantees (2) unconditionally, for any host and any contact solve, up to an accounting tolerance; the realized margins sit at the roundoff floor in all 90 measured cells (78 distinct configurations).
- (3) **The cost of enforcement.** Post-projection contact validity (penetration, corrective impulse, multiplier variance, momentum), measured rather than assumed (§3.3).

Relation to concurrent energy-control work. Wei et al. [15] guarantee discrete passivity at any finite inner-iteration budget via Douglas–Rachford splitting in wave coordinates, but their interfaces are *bilateral* couplings, not unilateral contact, and the guarantee is structural rather than a budgeted transfer cap. You et al. [16] modify implicit Euler to track user-specified *total-energy* targets bidirectionally, with barrier contact. Ours is neither: a one-directional cap on what contact may deposit into modal storage, funded by measured gross rigid-side loss and enforced by scaling realized modal state. Zobel et al. [18] solve ray-traced penalty contact quasi-statically without a shared unilateral multiplier; modes and modal derivatives have served as a *deformation* surrogate in a position-based host [13], whereas here modal amplitudes enter as contact DOFs driven by the shared multiplier.

Relation to passivity control and energy projection. The governor transplants an established control idea rather than inventing one. Time-domain passivity control [7] damps once an energy observer goes negative; energy-tank formulations [5] give that observer explicit state, so a controller spends only what its tank holds; Rath [14] enforces exact per-interface energy stability for contacting modal objects at audio rate, where we cap a scene-level scalar reservoir. We keep that structure and change two things: the tank is funded by *measured gross rigid-side loss*, not by the port it regulates, and the actuator scales realized *state* rather than modulating a force, because a fixed-budget solver commits its multipliers before the excess is observable. FEPR [4] also projects state post-solve, but onto *total-energy* conservation; ours is one-sided and never restores energy the solver lost.

2 The row and the cumulative bound

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law. It is one row *law*, instantiated at every support contact every substep — 48 such rows on the shelf, 40 on the ledge, 200 on the table — all sharing the one modal state $\mathbf{q}$.

The invariant. We require that the energy stored in the modal subsystem never exceed the *measured gross rigid-side loss* — defined exactly at (3) below — cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1]. \quad (2)$$

The transfer efficiency η sets how much of the measured loss may be spent; *every experiment in this paper runs* $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total remains within the budget, which is what lets a legitimate ring persist across substeps.

Definitions. Modal storage is the mechanical energy of the reduced state as written above, and $E_{\text{mod}}^0 = 0$: every scene starts from an undeformed, resting support, so even the run’s own settling transient must be funded by the reservoir. The supply is

$$\Delta E_{\text{rig}}^k = (E_{\text{rig}}^{k,-} - E_{\text{rig}}^{k,+}) + W_g^k, \quad W_g^k = \sum_b m_b \mathbf{g}(\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}), \quad (3)$$

where $E_{\text{rig}} = \sum_b [\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2} \omega_b^\top \mathbf{I}_b \omega_b]$ is *purely kinetic* — translational and rotational, over all dynamic bodies, not merely those in contact — and $\pm$ denote the two ends of substep k . We call this the **measured gross rigid-side loss**: the scene-wide, gravity-corrected, gross kinetic-energy loss of the rigid subsystem. It is a *loss*, not a dissipation — it also counts legitimate rigid-to-modal transfer and losses at rigid–rigid contacts elsewhere in the scene — so it is an envelope on what contact dissipated, not a measurement of it (§4). Gravity enters only through W_g , so free ballistic motion registers zero supply — a falling body’s kinetic gain exactly equals the gravity work, and only an inelastic velocity change funds the reservoir. Substep-boundary endpoints tile the timeline exactly, so no rigid energy change escapes the accounting.

Two objects, stated once. Conflating them is the easiest way to misread what follows. **(i) The printed invariant** is (2) exactly as written: a cumulative, gross-loss-funded storage ceiling on the modal subsystem whose supply is *measured, not prescribed* — the right-hand side is the measured loss of (3), never a tuned constant or a fraction of incident energy. It is not contact-port passivity and not a signed per-interface transfer bound: it constrains a scalar reservoir, not a port. The gross sum ignores substeps in which the rigid subsystem *gains* energy, so those cannot cancel earlier credits, and energy returned and re-dissipated counts as fresh supply (§4 measures both). **(ii) The enforced recursion** is the per-substep loop below — credit, test, project, debit — and it maintains (2) exactly,

not a relaxation of it: because the reservoir is credited *before* the projection test, the same-substep deposit that funds a legitimate rise in modal energy is already inside the budget the test enforces, so no granularity slack is required. Proposition 2.1 proves this; the only give is a per-substep accounting tolerance, which the governed runs realize at the 10^{-13} J roundoff floor.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage. Both happen *after* the contact solve, within the same substep: ΔE_{rig} is not observable until the velocity solve has finalized the rigid velocities, so the substep credits $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ at that point, tests the realized modal energy against it, and only then debits $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$. If the realized modal energy would overdraw B , the state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma(\mathbf{q}, \dot{\mathbf{q}}), \quad \gamma = \min\left(1, \sqrt{\frac{E_{\text{mod}}^- + B}{E_{\text{mod}}^+}}\right), \quad (4)$$

with $E_{\text{mod}}^\mp$ the modal energy before and after the solve. One substep of the governed loop, identical in all three hosts, then reads: (1) snapshot E_{rig}^- (pure KE, all bodies), $E_{\text{mod}}^-, \mathbf{x}^-$; (2) predict (gravity), generate contacts at the predicted pose; (3) position solve (K iterations), then velocity solve — the contact solve; (4) form $\Delta E_{\text{rig}} = (E_{\text{rig}}^- - E_{\text{rig}}^+) + W_g$, i.e. (3), post-solve; (5) credit $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$, same substep; (6) if $E_{\text{mod}}^+ > E_{\text{mod}}^- + B$, scale the state by γ of (4); (7) debit $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$ — contact is **not** re-solved.

The form (4) is closed because E_{mod} is quadratic in the state: scaling $\mathbf{q}$ and $\dot{\mathbf{q}}$ together scales both the kinetic and the elastic term by γ^2 , so γ bounds the total energy rather than the velocity alone. Two consequences are measured below: the projection is inert whenever the solve is within budget ($\gamma = 1$, bitwise-identical trajectories), and it never re-solves contact (step 7).

PROPOSITION 2.1. *For any $\eta \in (0, 1]$ and any host, the loop maintains $B \geq 0$ and guarantees (2) unconditionally, assuming nothing about the contact solve, up to an accounting tolerance ϵ over n substeps (ϵ the per-test tolerance, 10^{-12} J).*

Proof. Write $c_k = \eta \max(\Delta E_{\text{rig}}^k, 0)$ for the substep credit and B_k for the reservoir after substep k , and carry two invariants: $B_k \geq 0$ and $(E_{\text{mod}}^k - E_{\text{mod}}^0) + B_k \leq \sum_{j \leq k} c_j$. Both hold at $k=0$ ($B_0 = E_{\text{mod}}^0 = 0$). The credit (step 5) enters B *before* the test, so step 6 applies the largest γ with $\gamma^2 E_{\text{mod}}^+ \leq E_{\text{mod}}^- + B_{k-1} + c_k$; as E_{mod} is quadratic in the state, realized storage is at most $E_{\text{mod}}^- + B_{k-1} + c_k$, i.e. the realized gain $\Delta E_{\text{mod}} \leq B_{k-1} + c_k$. The debit (step 7) sets $B_k = \max(B_{k-1} + c_k - \max(\Delta E_{\text{mod}}, 0), 0)$, keeping $B_k \geq 0$; and in either branch $(E_{\text{mod}}^k - E_{\text{mod}}^0) + B_k \leq (E_{\text{mod}}^{k-1} - E_{\text{mod}}^0) + B_{k-1} + c_k$, so the second invariant follows by induction. Since $B_n \geq 0$, this is $E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \sum_{k \leq n} c_k$ — (2) itself, not a relaxation: the same-substep credit absorbs the granularity term a naive telescoping would otherwise leave. The 90 governed cells confirm it at the roundoff floor, worst margin 1.1×10^{-13} J.

3 Results

All solver-behaviour measurements below come from a single machine (Table 1), the sole exception being the device-resident timings of §3.5 (NVIDIA RTX 4090), reported separately; every number is frozen with its generating command and commit hash in the supplemental ledger. We deliberately avoid mixing machines: chaotic

Table 1: Host differences (above the rule) and settings held identical (below it). Entries are read from the implementation, with source anchors in the supplemental ledger.

	position-based	augmented-Lagr.	seq. impulse
unknown solved for	position	position + multiplier	velocity
contact treatment	compliant projection, $\tilde{\alpha} = \alpha/h^2$ ($\alpha=0$: rigid)	penalty + dual update, $\alpha=0.99$, $\beta=10^5$	Schur system on $\mathbf{JM}^{-1}\mathbf{J}^\top$, CFM-regularized
modal weight	explicit; per-mode compliance $1/(H_{ii}h^2 - 1)$	block-GS, full Newton step on $\mathbf{q}$	<i>implicit</i> , $(\mathbf{M} + h\mathbf{D} + h^2\mathbf{K})^{-1}$
iteration	GS sweeps, gap re-evaluated each sweep	coloured primal GS + $\mathbf{q}$ -block	PGS over rows
warm start	none ($\lambda \leftarrow 0$ each substep)	λ and penalty carried over	λ cache keyed per row
relaxation	0.7 on $\mathbf{q}$ and the support block	0.7 on the $\mathbf{q}$ -block Newton step	<i>inert</i> : never read
h ; substep	1/120 s; h/S , contacts regenerated per substep		
modal rank r	16 (shelf), 16 (ledge), 24 (table)		
modal damping	Rayleigh $\mathbf{D} = \alpha_0\mathbf{M} + \alpha_1\mathbf{K}$, $\alpha_0 \in \{2, 3\}$, $\alpha_1 = 10^{-5}$		
stepper; η	implicit midpoint; $\eta = 1$		
machine	Apple M4, CPU only, CPython 3.12		

contact stacks diverge across architectures under floating-point reassociation.

3.1 One row, three implementations, the same 24 cells

We drive an impactor into a modally reduced support in three scenes — a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [3] — and sweep modal relaxation and the local budget (iterations $\times$ substeps) over $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$. The hosts are XPBD [11], AVBD [6] (a descendant of vertex block descent [2]), and a sequential-impulse realization in the sense of Catto [1], made implicit in the modal block. Table 1 lists what differs and what is pinned identical. Two entries deserve emphasis. Relaxation is *inert* on the impulse host — its implicit modal weight needs no under-relaxation and the attribute is never read, hence the bit-identical hatched rows of Fig. 2. Warm start also differs, and we run each host as published: $\lambda \leftarrow 0$ every substep is the position-based host's own form and carried duals with a growing penalty are constitutive of the augmented-Lagrangian method, so neither is our choice. The comparison is as-deployed, not compliance-matched.

Equal row evaluations, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis spans $\times 64$ and conflates two refinements that are neither interchangeable nor equally priced (a substep also repeats contact generation and integration). At 32 row evaluations on the shelf drop, spent as iterations ($K=32, S=1$) the ratio is 0.300 and (2) holds; spent as substeps ($K=4, S=8$) it is 3.13, overdrawing by 481 J: substep refinement alone does not converge the row.

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation

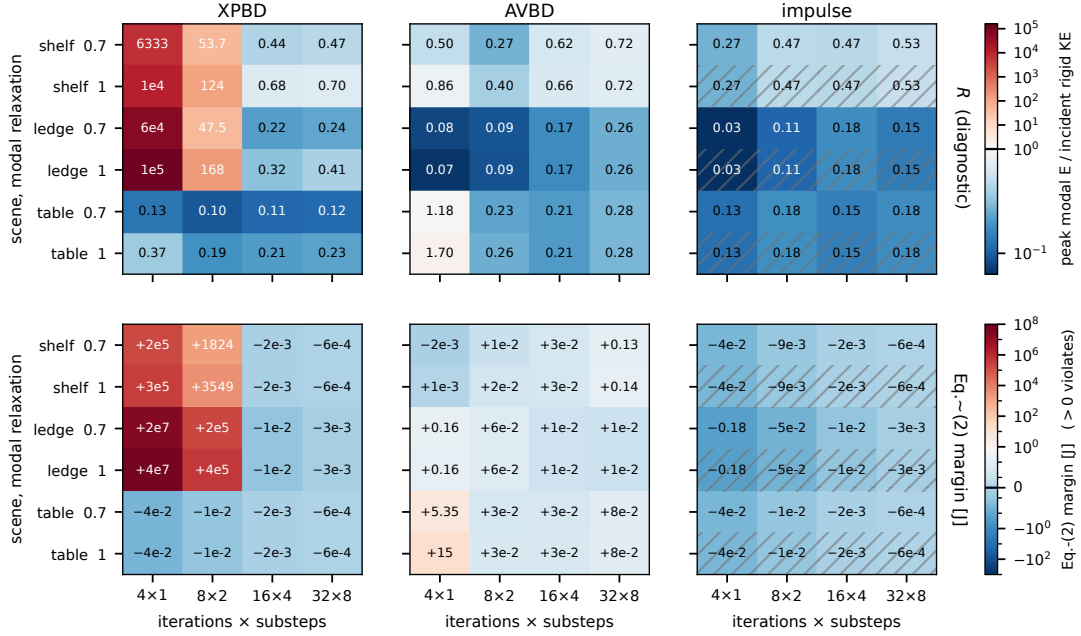


Figure 2: The same contact row in three fixed-budget solver implementations over an identical 24-cell sweep (3 scenes × modal relaxation {0.7, 1.0} × budget), governor OFF. Top: the ratio $R = \text{peak modal energy} / \text{peak incident rigid kinetic energy}$, log scale diverging at 1 — a severity diagnostic. XPBD exceeds 1 in 8/24 cells, AVBD in 2/24, the implicit realization in 0/24. Bottom: the invariant itself — the signed margin of (2) in joules, symmetric-log, diverging at 0; positive values violate. The two panels disagree on AVBD. Hatched rows mark a relaxation axis the host does not consume, bit-identical by construction (§3.1).

1.0, 4×1: peak modal energy 4.44×10^7 J against an incident 371 J; the augmented-Lagrangian host in 2 of 24 — both at the starved 4×1 corner of the table scene, worst 1.70 — so it is not unconditionally benign either. The ordering is scene-dependent and it inverts: worst-over-scenes at 4×1 the position-based host is five orders of magnitude more energetic (1.2×10^5 vs 1.70), yet on the table scene itself the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

The ratio is not the invariant. An impactor’s incident kinetic energy is not the supply (2) budgets against, so we measure (2) directly on all three hosts, ungoverned, running the reservoir accounting live while forcing $\gamma \equiv 1$: every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an ungoverned run (the ratios above reproduce the independent sweep exactly). We report the signed margin of (2) in joules; positive values violate. The invariant and the ratio do not flag the same cells: the position-based host violates (2) in 9 of 24 cells against $R > 1$ in 8, the augmented-Lagrangian host in 3 against 2, and the implicit host in neither (0/24, worst margin at the roundoff floor). Each metric catches a cell the other misses, so neither subsumes the other. What orders the failures is magnitude: the position-based overdraw reaches 4.4×10^7 J, five orders past the largest augmented-Lagrangian one (6.7 J, at the starved 4×1 table corner where $R = 1.70$), itself past the implicit host’s exact zero — three regimes the ratio-to-incident-energy diagnostic compresses into a single number.

With enforcement enabled, all 72 cells of this sweep satisfy (2) — 60 of them distinct configurations, the implicit host’s 12 relaxation pairs being bit-identical (Fig. 2); the 18 deployed-budget cells of §3.2 satisfy it too. The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell, nor at 16×4 or 32×8 on the position-based host at relaxation 0.7: it is inert precisely where the solve is already within budget.

3.2 The position-based amplification is a truncation artifact

Figure 3 isolates the local iteration budget. Two references recur below and are not the same object: *the implicit reference* is the implicit high-iteration realization at $K=500$ — the same code path run to $K=500$ — and *the XPBD self-reference* is the position-based host run to $K=500$ against itself. The position-based host falls monotonically from 2.96×10^4 at $K=1$ to 0.300 at $K=32$, approaching the implicit reference 0.2735; the gap $|\text{XPBD} - \text{reference}|$ shrinks from 2.96×10^4 to 2.7×10^{-2} . The implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (max–min 6.8×10^{-4}).

A falling energy ratio does not by itself show that the row is converging — a single scalar could shrink coincidentally. Measuring the constraint directly resolves the question, and the two complementarity conditions must be read in their own units rather than compounded into one dimensionally mixed scalar. End-of-substep penetration falls from 27.9 mm at $K=1$ to 3.6 μm by $K=64$, where it saturates. The multiplier surviving on *separated* rows — force where the gap is open, in units of the steady-state median — instead

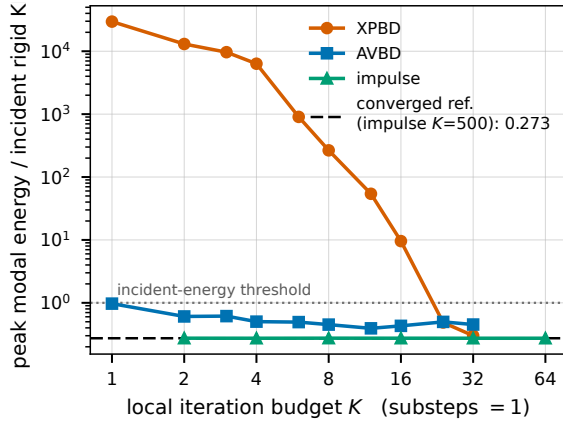


Figure 3: Energy ratio versus local iteration budget K on a deterministic shelf drop, substeps pinned at 1 so K is the only variable, governor OFF. The position-based host decays monotonically toward the implicit reference, crossing the incident-energy threshold between $K=16$ and $K=24$; the gap shrinks by six orders of magnitude. The reference is the implicit realization at $K=500$ — the same code path, not a different model; it is already converged at $K=2$. The AVBD curve is shown for completeness only: its denominator is itself budget-dependent at small K , so it cannot be read as a convergence trend.

rises from $84\times$ to $139\times$ through $K=24$, collapsing to exactly zero only at $K=64$. The energy crossing at $K \approx 24$ therefore coincides with the gap converging, not with complementarity holding: the multiplier side clears a rung later. Position-based host only — the augmented-Lagrangian multiplier is not the same object, and the implicit realization is converged at $K=2$.

Self-convergence stops short of the reference. Running the position-based host to $K=500$ against itself does not close the gap: its ratio plateaus by $K \approx 64$ at 0.2996 , so the residual 2.6×10^{-2} against the reference’s 0.2735 is a fixed point, not a tail. The state agrees no better — on the deflection trajectory $d_i(t) = U_y[i]^T q(t)$ the peak matches to 2.4% but the trajectories differ by 6.8 mm, 34% of the reference’s own peak. The two hosts converge to different discrete solutions, as expected when they discretize the same continuous law differently: the sweep thus separates a truncation pathology that convergence removes from a discretization difference that it does not.

This cuts two ways: truncation creates the amplification and convergence largely cures it, so iterating further — or adopting the implicit realization — would remove it outright, and the bound of §2 is not a substitute for convergence but targets the regime where neither escape is available. Following the one-iteration small-step regime [12], interactive position-based solvers run schedules near 1×8 to 2×4 , well below the $K \geq 24$ this scene needs. On 18 further cells (3 scenes $\times$ 3 hosts $\times$ $\{1\times 8, 2\times 4\}$, relaxation 0.7) the position-based host violates (2) in all six of its deployed cells, worst $R = 2282$ — on 8 row evaluations, twice the 4 of the 4×1 corner — while the others behave as in the sweep (1/6 and 0/6, the former by at

Table 2: Post-projection contact validity for the *position-based* host at starved budgets, relaxation 0.7, per clamp-active substep. Gap violation is measured immediately after the γ -scale; the corrective impulse is the next substep’s normal impulse against the unclamped steady-state median, a ratio because the hosts’ multipliers are different objects. The 4×1 cells clamp on nearly every substep, so no steady state exists there (—).

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
shelf	4×1	107/108	1.24 / 21.6	—	—
shelf	8×2	171/216	0.21 / 6.43	$3.68\times$	$5.6\times$
ledge	4×1	97/108	0.08 / 21.2	$1.03\times$	$1.08\times$
ledge	8×2	178/216	0.07 / 4.08	$8.92\times$	$62\times$

most 0.011 J). Governed, all 18 satisfy (2) (worst $R = 1.023$) and the projection is gentler: worst penetration 9.8 mm against 21.6.

Not a knife-edge configuration. Varying one setting at a time away from the two worst cells — h , contact and support compliance, modal rank, both Rayleigh coefficients — (2) is violated with $R > 1$ in all 24 configurations, R spanning 2.5 to 4.7×10^5 . Rank localizes the effect without removing it: excluding the stiff cluster outright (shelf $r=10$) cuts R from 6333 to 22.1, yet 584 J is still overdrawn. Convergence cures the worst cell as it does the shelf drop: at ledge relaxation 1.0, R falls $8.5 \times 10^5 \rightarrow 0.165$ over $K=1 \dots 32$.

3.3 The cost of enforcement

Accuracy. Bounding the energy is useful only if the bounded solution improves on the unbounded one. At a moderate injecting cell (shelf, 8×2 , relaxation 0.7), peak modal energy is 1555.6 J ungoverned and 29.56 J governed against the implicit reference’s 7.92 J: the energy error falls $196\times \rightarrow 3.7\times$ ($+1548 \rightarrow +22$ J), and $189\times \rightarrow 3.6\times$ against the XPBD self-reference (8.22 J), so it does not depend on the reference chosen. The trajectory, however, degrades: $\|d - d^{\text{ref}}\|_\infty$ is 6.5 mm ungoverned and 14.2 mm governed, 33% and 71% of the reference’s peak — the governor is a safety envelope, not an accuracy device. At the deployed 1×8 cell energy gives $2823\times \rightarrow 3.7\times$ and the trajectory improves too ($108 \rightarrow 96\%$ of reference peak): the trade is budget-dependent, but 96% is still not faithful. The 500×1 reference is converged in iteration count, not schedule-matched to the governed $S>1$ runs, so the residual bounds the governed error rather than isolating truncation from schedule mismatch.

This also explains how a $196\times$ energy error can coexist with a peak deflection only $1.2\times$ too large: the excess is neither kinetic (0.3%) nor low-frequency — the shelf spectrum is bimodal (ten bending modes below 2.1 kHz, six stiff above 20 kHz) and 99.6% of the ungoverned energy sits in the stiff cluster, $\sim 200\times$ the substep rate, against 0.0% for both references. A single scalar cannot redistribute energy across the spectrum: γ corrects the *amount* ($1555.6 \rightarrow 29.3$ J), leaves the character intact ($99.6 \rightarrow 97.7\%$), and scales the legitimate sag down with the noise ($24.1 \rightarrow 5.9$ mm against 20.0).

Contact validity. Since $\gamma < 1$ shrinks the sag $U_y^T q$, (4) lifts the support *into* the resting body without re-solving contact. Table 2 quantifies the consequence, instrumented by wrappers that leave

the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.24 mm, but the worst case reaches 21.6 mm, a 3.1–12.5× increase over its pre-scale value. The shelf board is 30 mm thick, so that is 72% of its thickness; against the unclamped scene at either reference (resting sag 1.7–2.4 mm, peak dynamic deflection 20.0–20.5 mm) it is 9–13× the sag and 1.05–1.08× the board's peak deflection: a visible interpenetration where the projection is most load-bearing. Contact recovers in the next substep rather than accumulating, but that correction costs a normal impulse of up to 8.9× the steady-state value, with multiplier variance up to 62× higher. Net rigid linear momentum is unchanged across the projection — exactly zero drift, verified numerically — in every clamp-active substep of every cell, necessarily so since (4) writes only modal state.

On the augmented-Lagrangian host the same instrumentation fires only at its one materially-overdrawing cell (the starved 4×1 table corner, +6.7 J), where the projection is an order of magnitude gentler: it opens 1.8 mm against the position-based host's 21.6, clamps a single substep against its near-continuous firing, and needs no measurable corrective impulse. The hosts differ in the scale of the correction, not its character (full rows in the supplement).

3.4 The reduced response against a full-FEM reference

Bounding the energy is likewise useful only if the bounded response is correct. On the ledge scene we compare the reduced arm against an unreduced FEM solve on the *same* discrete operator (the reduced basis is eigsh($\mathbf{K}, \mathbf{M}$) of the operator the reference integrates), so discretization bias cancels. The reduced-to-reference peak-deflection ratio rises monotonically $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$ as h refines $1/120 \rightarrow 1/960$, plateauing near 0.9; the residual $\sim 10\%$ is $k=24$ mode truncation on a thick slab, not a coupling error — the ring frequency agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees too: peak deflection sits at the pedestal antinode, not under the impact, and the falloff tracks the reference at Spearman $\rho = 0.89$ with no fitted attenuation term, in the small-displacement regime.

3.5 Runtime cost

On the CPU host at 16×4, over ten repetitions of 100 frames, the baseline step costs 11.0–125.6 ms by scene and host, and the ledger adds 0.23–0.41 ms — 0.9–3.4% — on the four shelf and ledge configurations, where it resolves above run-to-run variance. On the table scene it does not: the overhead's per-repetition spread exceeds its mean in both hosts ($+1.35 \pm 2.08$ and $+0.04 \pm 1.43$ ms), so we report it only as bounded by ~ 1.4 ms. The percentage is the portable quantity: on a second machine absolute times differ by 3.5–4.6× while the percentages agree to within 0.02 points wherever both resolve. These baselines fall 1.3–15× short of a 120 Hz budget, though shelf and ledge meet it at 2×4 governed. The percentage is budget-dependent and larger at the deployed schedules: at 1×8 and 2×4 the ledger's absolute cost stays comparable (0.30–2.22 ms) but the smaller baseline lifts it to 6.6–34.3% (shelf and ledge, both hosts), so enforcement is not free where the budget is tight.

A separate device-resident path (NVIDIA RTX 4090), carrying the ledger read-only as a monitor rather than enforcing it, reaches

5.3–9.4 ms/step at 16×4 across four scenes: shelf and ledge meet a 120 Hz budget with margin, a road slab sits at the boundary, the table does not, and all four meet it at 16×2 or below (per-scene factors in the supplement). We make no unqualified real-time claim, and a device-resident *enforced* γ does not exist in this work, so governed device timings lie outside our measurements (§4).

4 Limitations

Stability is not accuracy. Where the projection is load-bearing it opens contact violations up to 21.6 mm and moves the trajectory away from the implicit reference even as it moves the energy toward it (§3.3): bounded, not faithful, so its honest use is as a safety envelope for budgets that would otherwise diverge, and convergence remains the better remedy wherever affordable. A natural next mechanism scales deviations about the quasi-static equilibrium rather than the whole state, preserving the load-bearing sag; it is unbuilt.

Scope of the row. The coupling is normal-only — friction stays body-local, so scraping and rolling transfer lie outside it — and the position-level form fixes $e = 0$. The bound governs modal storage only, not energy created spuriously in rigid degrees of freedom: in the impulse realization, box–box rows rectifying a support ring into net rigid motion, to which the ledger is blind by design.

The supply is a gross sum, so it can be recycled. Energy returning from the modes to the rigid bodies and dissipated there is credited twice. We bound the exposure by accumulating the opposite channel, $\sum_k \max(-\Delta E_{\text{rig}}^k, 0)$, as a fraction of the supply: 0.4–27% (implicit), 3–32% (position-based) — larger than the $\leq 1\%$ we expected — and 102–118% on the augmented-Lagrangian host, whose rigid subsystem gains more than it loses in every cell, the box–box rectification above rather than a property of the bound. Tightening the envelope needs the signed or per-interface accounting this scalar reservoir avoids. The supply also depends on the substep *partition*: a finer schedule rectifies more fluctuation into supply, so the budget belongs to the schedule as well as the scene.

Coverage. Enforcement is host-side and single-machine (CPU float64); a device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work. Three formulations, one implementation of each; three scenes, one contact regime; the augmented-Lagrangian host still overdraws in 3 of 24 cells, so any claim that descent-class solvers are passive in this setting is an empirical statement about this sweep, not a theorem; since the implicit host's margins sit at exactly zero under the same accounting, that overdraft (up to 6.7 J) is a property of the solver, not the ledger.

5 Conclusion

Transcribing an established rigid-modal contact law into fixed-budget solvers is not energy-safe, and how unsafe differs sharply across the three tested implementations: worst-case modal-to-incident ratios are 1.2×10^5 , 1.70 and 0.53 on the same 24 cells. The position-based amplification is a truncation artifact convergence removes; the augmented-Lagrangian overdraft we measure but do not explain. Where convergence is unaffordable, a cumulative storage bound funded by measured gross rigid-side loss holds in every measured cell, at a quantified cost to contact validity.

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